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CHAPTER 5

Sterilization

INTRODUCTION

A FERMENTATION product is produced by the culture of a certain organism, or organisms, in a nutrient medium. If the fermentation is invaded by a foreign micro-organism then the following consequences may occur:

(i) The medium would have to support the growth of both the production organism and the contaminant, resulting in a loss of productivity.
(ii) If the fermentation is a continuous one then the contaminant may 'outgrow' the production organism and displace it from the fermentation.
(iii) The foreign organism may contaminate the final product, e.g. single-cell protein where the cells, separated from the broth, constitute the product.
(iv) The contaminant may produce compounds which make subsequent extraction of the final product difficult.
(v) The contaminant may degrade the desired product; this is common in bacterial contamination of antibiotic fermentations where the contaminant would have to be resistant to the normal inhibitory effects of the antibiotic and degradation of the antibiotic is a common resistance mechanism, e.g. the degradation of β-lactam antibiotics by β-lactamase-producing bacteria.
(vi) Contamination of a bacterial fermentation with phage could result in the lysis of the culture.

Avoidance of contamination may be achieved by:

(i) Using a pure inoculum to start the fermentation, as discussed in Chapter 6.
(ii) Sterilizing the medium to be employed.
(iii) Sterilizing the fermenter vessel.
(iv) Sterilizing all materials to be added to the fermentation during the process.
(v) Maintaining aseptic conditions during the fermentation.

The extent to which these procedures are adopted is determined by the likely probability of contamination and the nature of its consequences. Some fermentations are described as 'protected' — that is, the medium may be utilized by only a very limited range of micro-organisms, or the growth of the process organism may result in the development of selective growth conditions, such as a reduction in pH. The brewing of beer falls into this category; hop resins tend to inhibit the growth of many micro-organisms and the growth of brewing yeasts tends to decrease the pH of the medium. Thus, brewing worts are boiled, but not necessarily sterilized, and the fermenters are thoroughly cleaned with disinfectant solution but are not necessarily sterile. Also, the precautions used in the development of inoculum for brewing are far less stringent than, for example, in an antibiotic fermentation. However, the vast majority of fermentations are not 'protected' and, if contaminated, would suffer some of the consequences previously listed. The approaches adopted to avoid contamination will be discussed in more detail, apart from the development of aseptic inocula which is considered in Chapter 6 and the aseptic operation and containment of fermentation vessels which are discussed in Chapters 6 and 7.

MEDIUM STERILIZATION

As pointed out by Corbett (1985), media may be sterilized by filtration, radiation, ultrasonic treatment, chemical treatment or heat. However, for practical reasons, steam is used almost universally for the sterilization of fermentation media. The major exception is the use of filtration for the sterilization of media for animal-cell culture — such media are completely soluble and contain heat labile components making filtration the method of choice. Filtration techniques will be considered later in this chapter. Before the techniques which are used for the steam sterilization of culture media are discussed it is necessary to discuss the kinetics of sterilization. The destruction of micro-organisms by steam (moist heat) may be described as a first-order chemical reaction and, thus, may be represented by the following equation:

$$-\mathrm{d}N/\mathrm{d}t = kN \qquad (5.1)$$

where N is the number of viable organisms present,
t is the time of the sterilization treatment,
k is the reaction rate constant of the reaction, or the specific death rate.

It is important at this stage to appreciate that we are considering the total number of organisms present in the volume of medium to be sterilized, *not* the concentration — the minimum number of organisms to contaminate a batch is one, regardless of the volume of the batch. On integration of equation (5.1) the following expression is obtained:

$$N_t/N_0 = \mathrm{e}^{-kt} \qquad (5.2)$$

where N_0 is the number of viable organisms present at the start of the sterilization treatment,
N_t is the number of viable organisms present after a treatment period, t.

On taking natural logarithms, equation (5.2) is reduced to:

$$\ln (N_t/N_0) = -kt \qquad (5.3)$$

The graphical representations of equations (5.1) and (5.3) are illustrated in Fig. 5.1, from which it may be seen that viable organism number declines exponentially over the treatment period. A plot of the natural logarithm of $N_t|N_0$ against time yields a straight line, the slope of which equals $-k$. This kinetic description makes two predictions which appear anomalous:

(i) An infinite time is required to achieve sterile conditions (i.e. $N_t = 0$).
(ii) After a certain time there will be less than one viable cell present.

Thus, in this context, a value of N_t of less than one is considered in terms of the probability of an organism surviving the treatment. For example, if it were predicted that a particular treatment period reduced the population to 0.1 of a viable organism, this implies that the probability of one organism surviving the treatment is one in ten. This may be better expressed in practical terms as a risk of one batch in ten becoming contaminated. This aspect of contamination will be considered later.

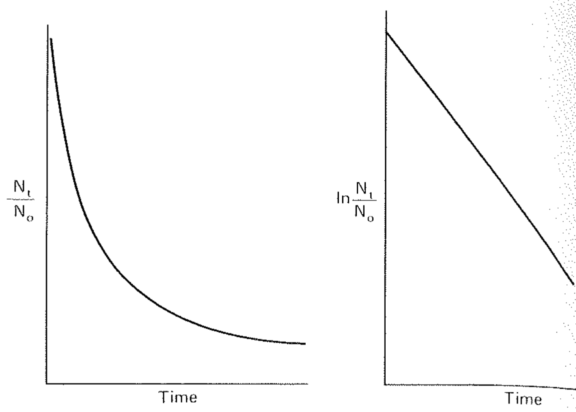

FIG. 5.1. Plots of the proportion of survivors and the natural logarithm of the proportion of survivors in a population of micro-organisms subjected to a lethal temperature over a time period.

The relationship displayed in Fig. 5.1 would be observed only with the sterilization of a pure culture in one physiological form, under ideal sterilization conditions. The value of k is not only species dependent, but dependent on the physiological form of the cell; for example, the endospores of the genus *Bacillus* are far more heat resistant than the vegetative cells. Richards (1968) produced a series of graphs illustrating the deviation from theory which may be experienced in practice. Figures 5.2a, 5.2b and 5.2c illustrate the effect of the time of heat treatment on the survival of a population of bacterial endospores. The deviation from an immediate exponential decline in viable spore number is due to the heat activation of the spores, that is the induction of spore germination by the heat and moisture of the initial period of the sterilization process. In Fig. 5.2a the activation of spores is significantly more than their destruction during the early stages of the process and, therefore, viable numbers increase before the observation of exponential decline. In Fig. 5.2b activation is balanced by spore death and in Fig. 5.2c activation is less than spore death.

Figures 5.3a and 5.3b illustrate typical results of the sterilization of mixed cultures containing two species with different heat sensitivities. In Fig. 5.3a the population consists mainly of the less-resistant type where the

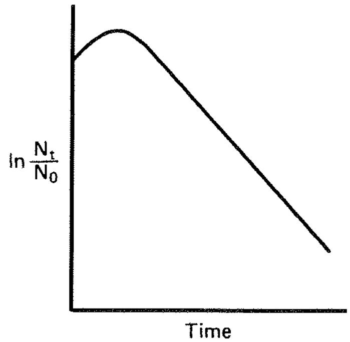

FIG. 5.2a. Initial population increase resulting from the heat activation of spores in the early stages of a sterilization process (Richards, 1968).

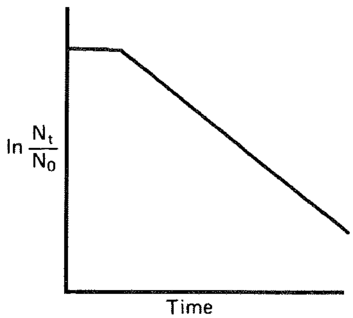

FIG. 5.2b. An initial stationary period observed during a sterilization treatment due to the death of spores being completly compensated by the heat activation of spores (Richards, 1968).

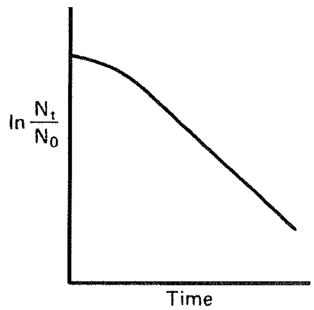

FIG. 5.2c. Initial population decline at a sub-maximum rate during a sterilization treatment due to the death of spores being compensated by the heat activation of spores (Richards, 1968).

initial decline is due principally to the destruction of the less-resistant cell population and the later, less rapid decline, is due principally to the destruction of the more resistant cell population. Figure 5.3b represents the reverse situation where the more resistant type predominates and its presence disguises the decrease in the number of the less resistant type.

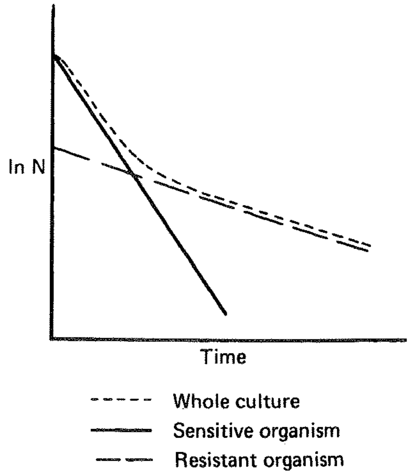

FIG. 5.3a. The effect of a sterilization treatment on a mixed culture consisting of a high proportion of a very sensitive organism (Richards, 1968).

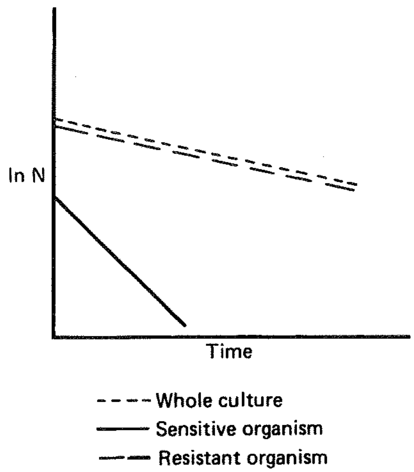

FIG. 5.3b. The effect of a sterilization treatment on a mixed culture consisting of a high proportion of a relatively resistant organism (Richards, 1968).

As with any first-order reaction, the reaction rate increases with increase in temperature due to an increase in the reaction rate constant, which, in the case of the destruction of micro-organisms, is the specific death rate (k). Thus, k is a true constant only under constant temperature conditions. The relationship between temperature and the reaction rate constant was demonstrated by Arrhenius and may be represented by the equation:

$$\mathrm{d}\ln k/\mathrm{d}T = E\,/\,RT^2 \qquad (5.4)$$

where E is the activation energy,
R is the gas constant,
T is the absolute temperature.
On integration equation (5.4) gives:

$$k = Ae^{-E/RT} \tag{5.5}$$

where A is the Arrhenius constant.

On taking natural logarithms, equation (5.5) becomes:

$$\ln k = \ln A - E/RT. \tag{5.6}$$

From equation (5.6) it may be seen that a plot of ln k against the reciprocal of the absolute temperature will give a straight line. Such a plot is termed an Arrhenius plot and enables the calculation of the activation energy and the prediction of the reaction rate for any temperature. By combining together equations (5.3) and (5.5), the following expression may be derived for the heat sterilization of a pure culture at a constant temperature:

$$\ln N_0/N_t = A \cdot t \cdot e^{-E/RT}. \tag{5.7}$$

Deindoerfer and Humphrey (1959) used the term ln N_0/N_t as a design criterion for sterilization, which has been variously called the Del factor, Nabla factor and sterilization criterion represented by the term ∇. Thus, the Del factor is a measure of the fractional reduction in viable organism count produced by a certain heat and time regime. Therefore:

$$\nabla = \ln (N_0/N_t)$$

but $$\ln(N_0/N_t) = kt$$

and $$kt = A \cdot t \cdot e^{-(E/RT)}$$

thus $$\nabla = A \cdot t \cdot e^{-(E/RT)}. \tag{5.8}$$

On rearranging, equation (5.8) becomes:

$$\ln t = E/RT + \ln (\nabla/A). \tag{5.9}$$

Thus, a plot of the natural logarithm of the time required to achieve a certain ∇ value against the reciprocal of the absolute temperature will yield a straight line, the slope of which is dependent on the activation energy, as shown in Fig. 5.4. From Fig. 5.4 it is clear that the same degree of sterilization (∇) may be obtained over a wide range of time and temperature regimes; that is, the same degree of sterilization may result from treatment at a high temperature for a short time as from a low temperature for a long time.

This kinetic description of bacterial death enables the design of procedures (giving certain ∇ factors) for the sterilization of fermentation broths. By choosing a value for N_t, procedures may be designed having a certain probability of achieving sterility, based upon the degree of risk that is considered acceptable. According to Deindoerfer and Humphrey (1959), Richards (1968), Banks (1979) and Corbett (1985) a risk factor of one batch in a thousand being contaminated is frequently used in the fermentation industry — that is, the final microbial count in the medium after sterilization should be 10^{-3} viable cells. However, to apply these kinetics it is necessary to know the thermal death characteristics of all the taxa contaminating the fermenter and unsterile medium. This is an impossibility and, therefore, the assumption may be made that the only microbial contaminants present are spores of *Bacillus stearothermophilus* — that is, one of the most heat-resistant microbial types known. Thus, by adopting *B. stearothermophilus* as the design organism a considerable safety factor should be built into the calculations. It should be remembered that *B. stearothermophilus* is not always adopted as the design organism. If the most heat-resistant organism contaminating the medium ingredients is known, then it may be advantageous to base the sterilization process on this organism. Deindoerfer and Humphrey (1959) determined the thermal death characteristics of *B. stearothermophilus* spores as:

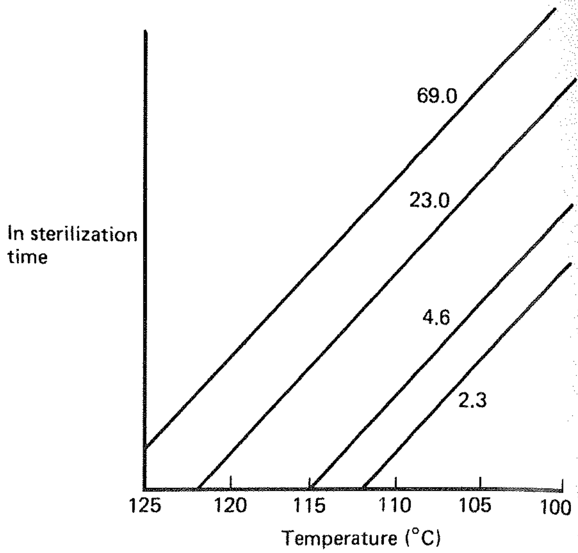

FIG. 5.4. The effect of sterilization and temperature on the Del factor achieved in the process. The figures on the graph indicate the Del factors for each straight line (modified after Richards, 1966).

$$\text{Activation energy} = 67.7 \text{ kcal mole}^{-1}$$

$$\text{Arrhenius constant} = 1 \times 10^{36.2} \text{ second}^{-1}$$

However, it should be remembered that these kinetic values will vary according to the medium in which the spores are suspended, and this is particularly relevant when considering the sterilization of fats and oils (which are common fermentation substrates) where the relative humidity may be quite low. Bader *et al.* (1984) demonstrated that spores of *Bacillus macerans* suspended in oil were ten times more resistant to sterilization if they were dry than if they were wet.

A regime of time and temperature may now be determined to achieve the desired Del factor. However, a fermentation medium is not an inert mixture of components, and deleterious reactions may occur in the medium during the sterilization process, resulting in a loss of nutritive quality. Thus, the choice of regime is dictated by the requirement to achieve the desired reduction in microbial content with the least detrimental effect on the medium. Figure 5.5 illustrates the deleterious effect of increasing medium sterilization time on the yield of product of subsequent fermentations. The initial rise in yield is due to some components of the medium being made more available to the process micro-organism by the 'cooking effect' of a brief sterilization period (Richards, 1966).

Two types of reaction contribute to the loss of nutrient quality during sterilization:

(i) *Interactions between nutrient components of the medium.* A common occurrence during sterilization is the Maillard-type browning reaction which results in discoloration of the medium as well as loss of nutrient quality. These reactions are normally caused by the reaction of carbonyl groups, usually from reducing sugars, with the amino groups of amino acids and proteins. An example of the effect of sterilization time on the availability of glucose in a corn-steep liquor medium is shown in Table 5.1 (Corbett, 1985). Problems of this type are normally resolved by sterilizing the sugar separately from the rest of the medium and recombining the two after cooling.

(ii) *Degradation of heat labile components.* Certain vitamins, amino acids and proteins may be degraded during a steam sterilization regime. In extreme cases, such as the preparation of media for animal-cell culture, filtration may be used and this aspect will be discussed later in the chapter. However, for the vast majority of fermentations these problems may be resolved by the judicious choice of steam sterilization regime.

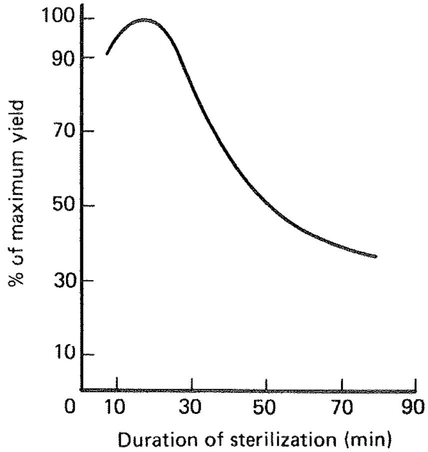

FIG. 5.5. The effect of the time of sterilization on the yield of a subsequent fermentation (Richards, 1966).

The thermal destruction of essential media components conforms approximately with first order reaction kinetics and, therefore, may be described by equations similar to those derived for the destruction of bacteria:

$$x_t/x_0 = e^{-kt} \tag{5.10}$$

where x_t is the concentration of nutrient after a heat treatment period, t,
x_0 is the original concentration of nutrient at the onset of sterilization,
k is the reaction rate constant.

It is important to appreciate that we are considering the decline in the concentration of the nutrient component, whereas we consider the decline in the number of contaminants. The effect of temperature on the reaction rate constant may be expressed by the Arrhenius equation:

$$\ln k = \ln A - E/RT.$$

Therefore, a plot of the natural logarithm of the reaction rate against $1/T$ will give a straight line, slope $-(E/R)$. As the value of R, the gas constant, is fixed

TABLE 5.1. *The effect of sterilization time on glucose concentration and product accretion rate in an antibiotic fermentation* (Corbett, 1985)

Time at 121° (min)	Amount of added glucose remaining (%)	Relative accretion rate
60	35	90
40	46	92
30	64	100

the slope of the graph is determined by the value of the activation energy (E). The activation energy for the thermal destruction of *B. stearothermophilus* spores has been cited as 67.7 kcal mole^{-1}, whereas that for thermal destruction of nutrients is 10 to 30 kcal mole^{-1} (Richards, 1968). Figure 5.6 is an Arrhenius plot for two reactions — one with a lower activation energy than the other. From this plot it may be seen that as temperature is increased, the reaction rate rises more rapidly for the reaction with the higher activation energy. Thus, considering the difference between activation energies for spore destruction and nutrient degradation, an increase in temperature would accelerate spore destruction more than medium denaturation.

In the consideration of Del factors it was evident that the same Del factor could be achieved over a range of temperature/time regimes. Thus, it would appear to be advantageous to employ a high temperature for a short time to achieve the desired probability of sterility, yet causing minimum nutrient degradation. Thus, the ideal technique would be to heat the fermentation medium to a high temperature, at which it is held for a short period, before being cooled rapidly to the fermentation temperature. However, it is obviously impossible to heat a batch of many thousands of litres of broth in a tank to a high temperature, hold for a short period and cool without the heating and cooling periods contributing considerably to the total sterilization time. The only practical method of materializing the objective of a short-time, high-temperature treatment is to sterilize the medium in a continuous stream. In the past the fermentation industry was reluctant to adopt continuous sterilization due to a number of disadvantages outweighing the advantage of nutrient quality. The relative merits of batch and continuous sterilization may be summarized as follows:

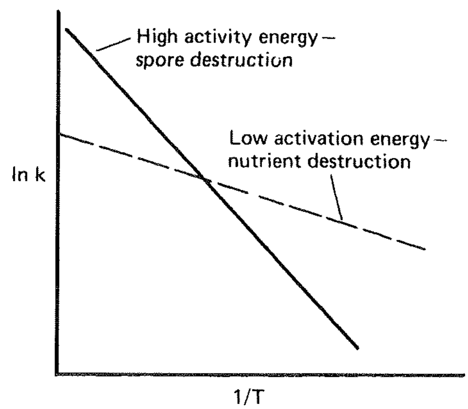

FIG. 5.6. The effect of activation energy on spore and nutrient destruction.

Advantages of continuous sterilization over batch sterilization

(i) Superior maintenance of medium quality.
(ii) Ease of scale-up — discussed later.
(iii) Easier automatic control.
(iv) The reduction of surge capacity for steam.
(v) The reduction of sterilization cycle time.
(vi) Under certain circumstances, the reduction of fermenter corrosion.

Advantages of batch sterilization over continuous sterilization

(i) Lower capital equipment costs.
(ii) Lower risk of contamination — continuous processes require the aseptic transfer of the sterile broth to the sterile vessel.
(iii) Easier manual control.
(iv) Easier to use with media containing a high proportion of solid matter.

The early continuous sterilizers were constructed as plate heat exchangers and these were unsuitable on two accounts:

(i) Failure of the gaskets between the plates resulted in the mixing of sterile and unsterile streams.
(ii) Particulate components in the media would block the heat exchangers.

However, modern continuous sterilizers use double spiral heat exchangers in which the two streams are separated by a continuous steel division. Also, the spiral exchangers are far less susceptible to blockage. However, a major limitation to the adoption of continuous sterilization was the precision of control necessary for its success. This precision has been achieved with the development of sophisticated computerized monitoring and control systems resulting in continuous sterilization being very widely used and it is now the method of choice. However, batch sterilization is still used in many fermentation plants and, thus, it will be considered here before continuous sterilization is discussed in detail.

THE DESIGN OF BATCH STERILIZATION PROCESSES

Although a batch sterilization process is less successful in avoiding the destruction of nutrients than a continuous one, the objective in designing a batch process is still to achieve the required probability of obtaining sterility with the minimum loss of nutritive quality. The highest temperature which appears to be feasible for batch sterilization is 121°C so the procedure should be designed such that exposure of the medium to this temperature is kept to a minimum. This is achieved by taking into account the contribution made to the sterilization by the heating and cooling periods of the batch treatment. Deindoerfer and Humphrey (1959) presented a method to assess the contribution made by the heating and cooling periods. The following information must be available for the design of a batch sterilization process:

(i) A profile of the increase and decrease in the temperature of the fermentation medium during the heating and cooling periods of the sterilization cycle.
(ii) The number of micro-organisms originally present in the medium.
(iii) The thermal death characteristics of the 'design' organism. As explained earlier this may be *Bacillus stearothermophilus* or an alternative organism relevant to the particular fermentation.

Knowing the original number of organisms present in the fermenter and the risk of contamination considered acceptable, the required Del factor may be calculated. A frequently adopted risk of contamination is 1 in 1000, which indicates that N_t should equal 10^{-3} of a viable cell. It is worth reinforcing at this stage that we are considering the total number of organisms present in the medium and *not* the concentration. If a specific case is considered where the unsterile broth was shown to contain 10^{11} viable organisms, then the Del factor may be calculated, thus:

$$\nabla = \ln\left(10^{11}/10^{-3}\right)$$

$$\nabla = \ln 10^{14}$$

$$= 32.2.$$

Therefore, the overall Del factor required is 32.2. However, the destruction of cells occurs during the heating and cooling of the broth as well as during the period at 121°C, thus, the overall Del factor may be represented as:

$$\nabla_{\text{overall}} = \nabla_{\text{heating}} + \nabla_{\text{holding}} + \nabla_{\text{cooling}}.$$

Knowing the temperature–time profile for the heating and cooling of the broth (prescribed by the characteristics of the available equipment) it is possible to determine the contribution made to the overall Del factor by these periods. Thus, knowing the Del factors contributed by heating and cooling, the holding time may be calculated to give the required overall Del factor.

Calculation of the Del factor during heating and cooling

The relationship between Del factor, the temperature and time is given by equation (5.8):

$$\nabla = A \cdot t \cdot e^{-(E/RT)}.$$

However, during the heating and cooling periods the temperature is not constant and, therefore, the calculation of ∇ would require the integration of equation (5.8) for the time–temperature regime observed. Deindoerfer and Humphrey (1959) produced integrated forms of the equation for a variety of temperature–time profiles, including linear, exponential and hyperbolic. However, the regime observed in practice is frequently difficult to classify, making the application of these complex equations problematical. Richards (1968) demonstrated the use of a graphical method of integration and this is illustrated in Fig. 5.7. The time axis is divided into a number of equal increments, t_1, t_2, t_3, etc., Richards suggesting 30 as a reasonable number.

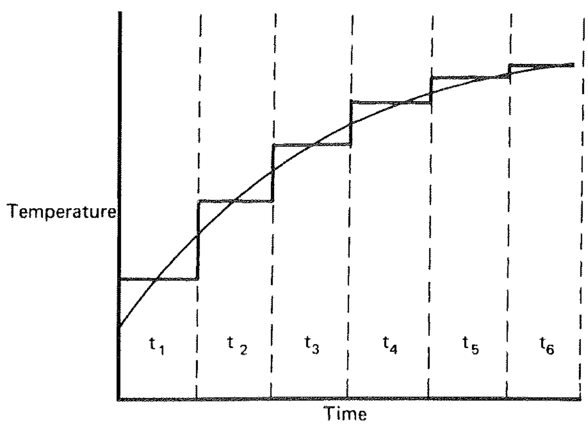

FIG. 5.7. The graphical integration method applied to the increase in temperature over a time period. t_1, t_2, etc. represent equal time intervals (Richards, 1968).

For each increment, the temperature corresponding to the mid-point time is recorded. It may now be approximated that the total Del factor of the heating-up period is equivalent to the sum of the Del factors of the mid-point temperatures for each time increment. The value of the specific death rate of *B. stearothermophilus* spores at each mid-point temperature may be deduced from the Arrhenius equation using the thermal death characteristic published by Deindoerfer and Humphrey (1959). The value of the Del factor corresponding to each time increment may then be calculated from the equations:

$$\nabla_1 = k_1 t,$$
$$\nabla_2 = k_2 t,$$
$$\nabla_3 = k_3 t,$$
$$\text{etc.}$$

The sum of the Del factors for all the increments will then equal the Del factor for the heating-up period. The Del factor for the cooling-down period may be calculated in a similar fashion.

Calculation of the holding time at constant temperature

From the previous calculations the overall Del factor, as well as the Del factors of the heating and cooling parts of the cycle, have been determined. Therefore, the Del factor to be achieved during the holding time may be calculated by difference:

$$\nabla_{\text{holding}} = \nabla_{\text{overall}} - \nabla_{\text{heating}} - \nabla_{\text{cooling}}$$

Using our example where the overall Del factor is 32.2 and if it is taken that the heating Del factor was 9.8 and the cooling Del factor 10.1, the holding Del factor may be calculated:

$$\nabla_{\text{holding}} = 32.2 - 9.8 - 10.1,$$
$$\nabla_{\text{holding}} = 12.3.$$

But $\nabla = kt$, and from the data of Deindoerfer and Humphrey (1961) the specific death rate of *B. stearothermophilus* spores at 121°C is 2.54 min^{-1}.

Therefore, $t = \nabla/k$ or $t = 12.3/2.54 = 4.84$ min.

If the contribution made by the heating and cooling parts of the cycle were ignored then the holding time would be given by the equation:

$$t = \nabla_{\text{overall}}/k = 32.2/2.54 = 12.68 \text{ min}.$$

Thus, by considering the contribution made to the sterilization process by the heating and cooling parts of the cycle a considerable reduction in exposure time is achieved

Richards' rapid method for the design of sterilization cycles

Richards (1968) proposed a rapid method for the design of sterilization cycles avoiding the time-consuming graphical integrations. The method assumes that all spore destruction occurs at temperatures above 100°C and that those parts of the heating and cooling cycle above 100° are linear. Both these assumptions appear reasonably valid and the technique loses very little in accuracy and gains considerably in simplicity. Furthermore, based on these assumptions, Richards has presented a table of Del factors for *B. stearothermophilus* spores which would be obtained in heating and cooling a broth up to (and down from) holding temperatures of 101–130°C, based on a temperature change of 1°C per minute. This information is presented in Table 5.2, together with the specific death rates for *B. stearothermophilus* spores over the temperature range. If the rate of temperature change is 1° per minute, the Del factors for heating and cooling may be read directly from the table; if the temperature change deviates from 1° per minute, the Del factors may be altered by simple proportion. For example, if a fermentation broth were heated from 100° to 121°C in 30 minutes and cooled from 121° to 100° in 17 minutes, the Del factors for the heating and cooling cycles may be determined as follows:

From Table 5.2, if the change in temperature had been 1° per minute, the Del factor for both the heating and cooling cycles would be 12.549. But the temperature change in the heating cycle was 21° in 30 minutes; therefore,

$$\text{Del}_{\text{heating}} = (12.549 \times 30)/21 = 17.93$$

and the temperature change in the cooling cycle was 21° in 17 minutes, therefore,

$$\text{Del}_{\text{cooling}} = (12.549 \times 17)/21 = 10.16.$$

Having calculated the Del factors for the heating and cooling periods the holding time at the constant temperature may be calculated as before.

The scale up of batch sterilization processes

The use of the Del factor in the scale up of batch sterilization processes has been discussed by Banks

TABLE 5.2. *Del values for* B. stearothermophilus *spores for the heating-up period over a temperature range of 100 to 130°, assuming a rate of temperature change of 1° min^{-1} and negligible spore destruction at temperatures below 100°* (Richards, 1968)

T (°C)	k (min^{-1})	∇
100	0.019	—
101	0.025	0.044
102	0.032	0.076
103	0.040	0.116
104	0.051	0.168
105	0.065	0.233
106	0.083	0.316
107	0.105	0.420
108	0.133	0.553
109	0.168	0.720
110	0.212	0.932
111	0.267	1.199
112	0.336	1.535
113	0.423	1.957
114	0.531	2.488
115	0.666	3.154
116	0.835	3.989
117	1.045	5.034
118	1.307	6.341
119	1.633	7.973
120	2.037	10.010
121	2.538	12.549
122	3.160	15.708
123	3.929	19.638
124	4.881	24.518
125	6.056	30.574
126	7.506	38.080
127	9.293	47.373
128	11.494	58.867
129	14.200	73.067
130	17.524	90.591

(1979). It should be appreciated by this stage that the Del factor does not include a volume term, i.e. absolute numbers of contaminants and survivors are considered, *not* their concentration. Thus, if the size of a fermenter is increased the initial number of spores in the medium will also be increased, but if the same probability of achieving sterility is required the final spore number should remain the same, resulting in an increase in the Del factor. For example, if a pilot sterilization were carried out in a 1000-dm^3 vessel with a medium containing 10^6 organisms cm^{-3} requiring a probability of contamination of 1 in 1000, the Del factor would be:

$$\nabla = \ln \{(10^6 \times 10^3 \times 10^3)/10^{-3}\}$$
$$= \ln (10^{12}/10^{-3})$$
$$= \ln 10^{15} = 34.5.$$

If the same probability of contamination were required in a 10,000-dm^3 vessel using the same medium the Del factor would be:

$$\nabla = \ln \{(10^6 \times 10^3 \times 10^4)/10^{-3}\}$$
$$= \ln (10^{13}/10^{-3})$$
$$= \ln 10^{16} = 36.8.$$

Thus, the Del factor increases with an increase in the size of the fermenter volume. The holding time in the large vessel may be calculated by the graphical integration method or by the rapid method of Richards (1968), as discussed earlier, based on the temperature–time profile of the sterilization cycle in the large vessel. However, it must be appreciated that extending the holding time on the larger scale (to achieve the increased ∇ factor) will result in increased nutrient degradation. Also, the contribution of the heating-up and cooling-down periods to nutrient destruction will be greater as scale increases. Maintaining the same nutrient quality on a small and a large scale can be achieved in batch sterilization only by compromising the sterility of the vessel, which is totally unacceptable. Thus, the decrease in the yield of a fermentation when it is scaled up is often due to problems of nutrient degradation during batch sterilization and the only way to eradicate the problem is to sterilize the medium continuously.

Methods of batch sterilization

The batch sterilization of the medium for a fermentation may be achieved either in the fermentation vessel or in a separate mash cooker. Richards (1966) considered the relative merits of *in situ* medium sterilization and the use of a special vessel. The major advantages of a separate medium sterilization vessel may be summarized as:

(i) One cooker may be used to serve several fermenters and the medium may be sterilized as the fermenters are being cleaned and prepared for the next fermentation, thus saving time between fermentations.
(ii) The medium may be sterilized in a cooker in a more concentrated form than would be used in the fermentation and then diluted in the fermenter with sterile water prior to inoculation. This would allow the construction of smaller cookers.
(iii) In some fermentations, the medium is at its most viscous during sterilization and the power requirement for agitation is not alleviated by

aeration as it would be during the fermentation proper. Thus, if the requirement for agitation during *in situ* sterilization were removed, the fermenter could be equipped with a less powerful motor. Obviously, the sterilization kettle would have to be equipped with a powerful motor, but this would provide sterile medium for several fermenters.

(iv) The fermenter would be spared the corrosion which may occur with medium at high temperature.

The major disadvantages of a separate medium sterilization vessel may be summarized as:

(i) The cost of constructing a batch medium sterilizer is much the same as that for the fermenter.

(ii) If a cooker serves a large number of fermenters complex pipework would be necessary to transport the sterile medium, with the inherent dangers of contamination.

(iii) Mechanical failure in a cooker supplying medium to several fermenters would render all the fermenters temporarily redundant. The provision of contingency equipment may be prohibitively costly.

Overall, the pressure to decrease the 'down time' between fermentations has tended to outweigh the perceived disadvantages of using separate sterilization vessels. Thus, sterilization in dedicated vessels is the method of choice for batch sterilization. However, as pointed out by Corbett (1985), the fact that separate batch sterilizers are used lends further weight to the argument that continuous sterilization should be adopted in preference to batch. The capital cost of a separate batch sterilizer is similar to that of a continuous one and the problems of transfer of sterile media are then the same for both batch and continuous sterilization. Thus, two of the major objections to continuous systems (capital cost and aseptic transfer) may be considered as no longer relevant.

THE DESIGN OF CONTINUOUS STERILIZATION PROCESSES

The design of continuous sterilization cycles may be approached in exactly the same way as for batch sterilization systems. The continuous system includes a time period during which the medium is heated to the sterilization temperature, a holding time at the desired temperature and a cooling period to restore the medium to the fermentation temperature. The temperature of the medium is elevated in a continuous heat exchanger and is then maintained in an insulated serpentine holding coil for the holding period. The length of the holding period is dictated by the length of the coil and the flow rate of the medium. The hot medium is then cooled to the fermentation temperature using two sequential heat exchangers — the first utilizing the incoming medium as the cooling source (thus conserving heat by heating-up the incoming medium) and the second using cooling water. The major advantage of the continuous process is that a much higher temperature may be utilized, thus reducing the holding time and reducing the degree of nutrient degradation. The required Del factor may be achieved by the combination of temperature and holding time which gives an acceptably small degree of nutrient decay. Richards (1968) quoted the following example to illustrate the range of temperature–time regimes which may be employed to achieve the same probability of obtaining sterility. The Del factor for the example sterilization was 45.7 and the following temperature time regimes were calculated to give the same Del factor:

Temperature	Holding time
130°	2.44 minutes
135°	51.9 seconds
140°	18.9 seconds
150°	2.7 seconds

Furthermore, because a continuous process involves treating small increments of medium the heating-up and cooling-down periods are very small compared with those in a batch system. There are two types of continuous sterilizer which may be used for the treatment of fermentation media: the indirect heat exchanger and the direct heat exchanger (steam injector).

The most suitable indirect heat exchangers are of the double-spiral type which consists of two sheets of high-grade stainless steel which have been curved around a central axis to form a double spiral, as shown in Fig. 5.8. The ends of the spiral are sealed by covers. A full scale example is shown in Fig. 5.9. To achieve sterilization temperatures steam is passed through one spiral and medium through the other in countercurrent

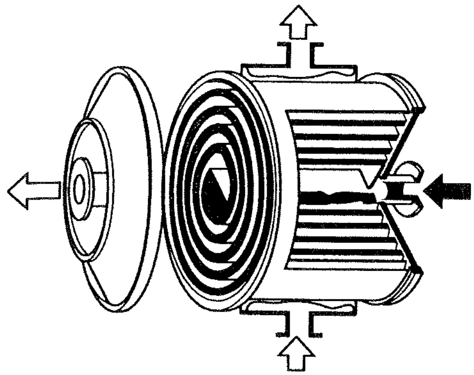

FIG. 5.8. A schematic representation of a spiral heat exchanger (Alfa-Laval Engineering Ltd, Brentford, Middlesex).

streams. Spiral heat exchangers are also used to cool the medium after passing through the holding coil. Incoming unsterile medium is used as the cooling agent in the first cooler so that the incoming medium is partially heated before it reaches the sterilizer and, thus, heat is conserved.

The major advantages of the spiral heat exchanger are:

(i) The two streams of medium and cooling liquid, or medium and steam, are separated by a continuous stainless steel barrier with gasket seals being confined to the joints with the end plates. This makes cross contamination between the two streams unlikely.
(ii) The spiral route traversed by the medium allows sufficient clearances to be incorporated for the system to cope with suspended solids. The exchanger tends to be self-cleaning which reduces the risk of sedimentation, fouling and 'burning-on'.

Indirect plate heat exchangers consist of alternating plates through which the countercurrent streams are circulated. The plates are separated by gaskets and failure of these gaskets can cause cross-contamination between the two streams. Also, the clearances between the plates are such that suspended solids in the medium may block the exchanger and, thus, the system is only useful in sterilizing completely soluble media. However, the plate exchanger is more adaptable than the spiral system in that extra plates may be added to increase its capacity.

The continuous steam injector injects steam directly into the unsterile broth. The advantages and disadvantages of the system have been summarized by Banks (1979):

(i) Very short (almost instantaneous) heating up times.
(ii) It may be used for media containing suspended solids.
(iii) Low capital cost.
(iv) Easy cleaning and maintenance.
(v) High steam utilization efficiency.

However, the disadvantages are:

(i) Foaming may occur during heating.
(ii) Direct contact of the medium with steam requires that allowance be made for condense dilution and requires 'clean' steam, free from anticorrosion additives.

In some cases the injection system is combined with flash cooling, where the sterilized medium is cooled by passing it through an expansion valve into a vacuum chamber. Cooling then occurs virtually instantly. A flow chart of a continuous sterilization system using direct steam injection is shown in Fig. 5.10. In some cases a combination of direct and indirect heat exchangers may be used (Svensson, 1988). This is especially true for starch-containing broths when steam injection is used for the pre-heating step. By raising the temperature virtually instantaneously the critical gelatinization temperature of the starch is passed through very quickly and the increase in viscosity normally associated with heated starch colloids can be reduced.

The most widely used continuous sterilization system is that based on the spiral heat exchangers and a typical layout is shown in Fig. 5.11. The plant is sterilized prior to sterilization of the medium by circulating hot water through the plant in a closed circuit. At the same time, the fermenter and the pipework between the fermenter and the sterilizer are steam sterilized. Heat conservation is achieved by cooling the sterile medium against cold, incoming unsterile medium which will then be partially heated before it reaches the sterilizer.

The Del factor to be achieved in a continuous sterilization process has to be increased with an increase in scale, and this is calculated exactly as described in the consideration of the scale up of batch regimes. Thus, if the volume to be sterilized is increased from 1000 dm^3 to 10,000 dm^3 and the risk of failure is to remain at 1 in 1000 then the Del factor must be increased from 34.5 to 36.8. However, the advantage of the continuous

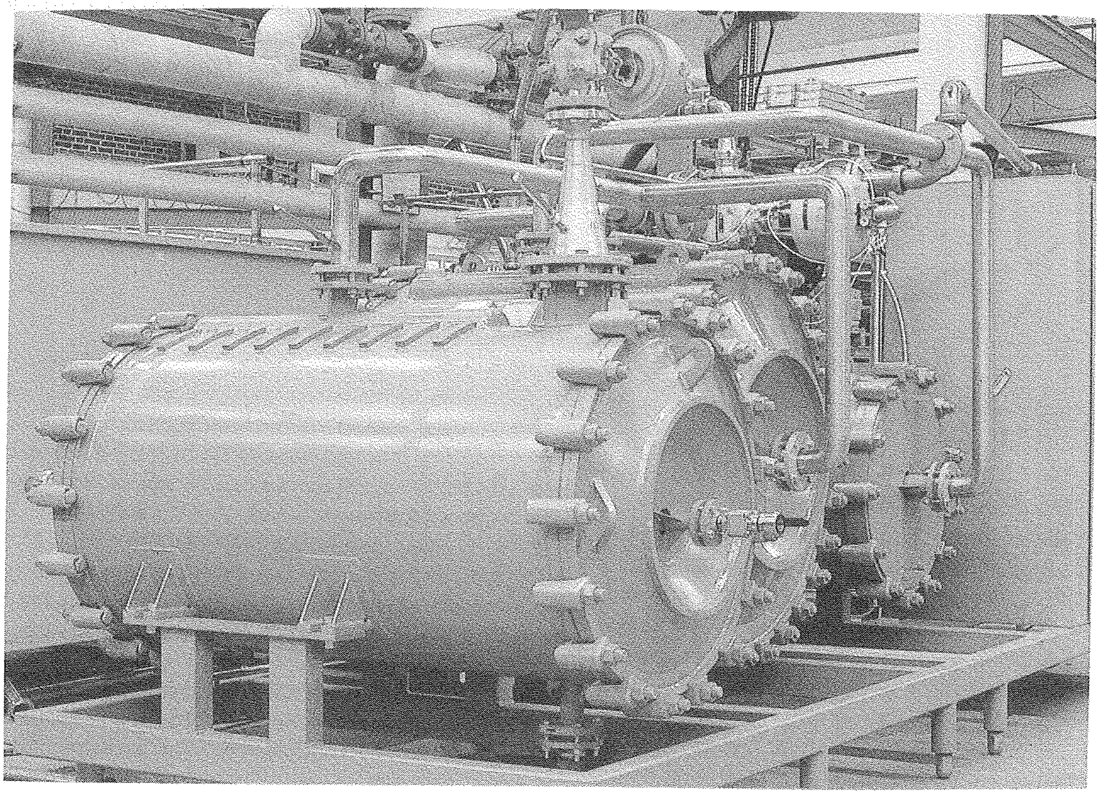

FIG. 5.9. Industrial scale spiral heat exchanger (Alfa-Laval Engineering Ltd., Brentford, Middlesex).

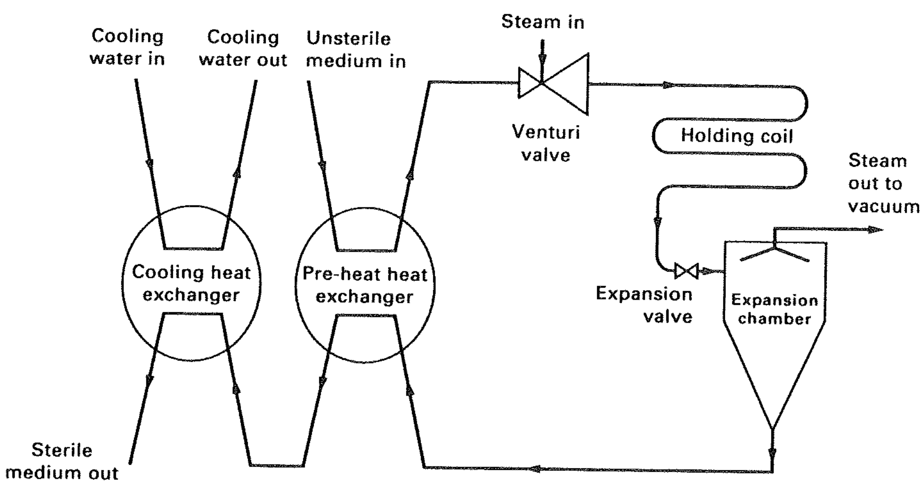

FIG. 5.10. Flow diagram of a typical continuous injector-flash cooler sterilizer.

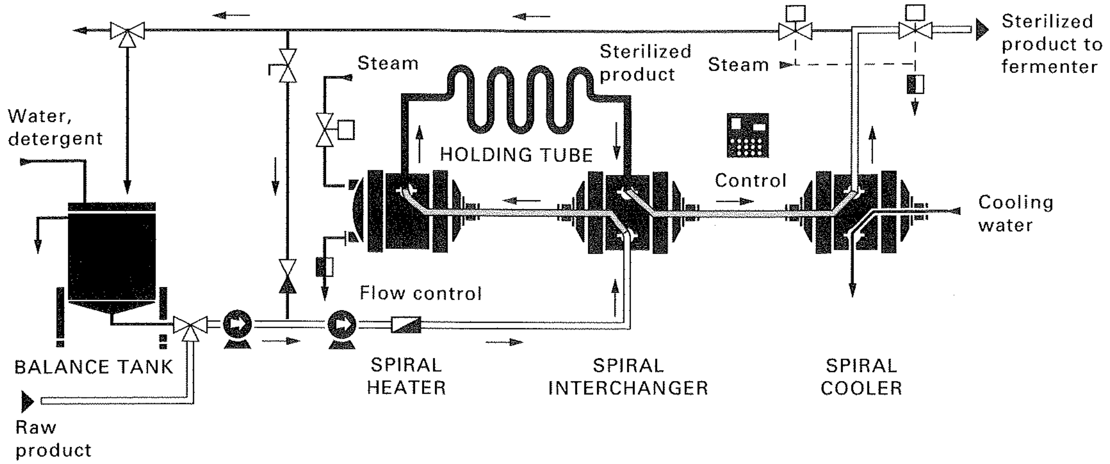

FIG. 5.11. Flow diagram of a typical continuous sterilization system employing spiral heat exchangers (Alfa-Laval Engineering Ltd., Brentford, Middlesex).

process is that temperature may be used as a variable in scaling up a continuous process so that the increased ∇ factor may be achieved whilst maintaining the nutrient quality constant. Deindoerfer and Humphrey (1961) attempted to rationalize the choice of time–temperature regime by the use of a Nutrient Quality Criterion (Q), based on similar logic to the Del factor:

$$Q = \ln (x_0/x_t) \tag{5.11}$$

where x_0 is the concentration of essential heat labile nutrient in the original medium,
x_t is the concentration of essential heat labile nutrient in the medium after a sterilization time, t.

As considered earlier, the destruction of a nutrient may be considered a first-order reaction:

$$x_t/x_0 = e^{-kt}$$

where k is the reaction rate constant

or $$x_0/x_t = e^{kt}.$$

Thus, taking natural logarithms,

$$\ln (x_0/x_t) = kt.$$

Therefore $$Q = kt.$$

The relationship between k and absolute temperature is described by the Arrhenius equation:

$$k = A \cdot e^{-(E/RT)}$$

therefore, substituting for k:

$$Q = A \cdot t \cdot e^{-(E/RT)}.$$

Therefore, as for the Del factor equation, by taking natural logarithms, and rearranging, the following equation is obtained

$$\ln t = E/RT + \ln Q/A. \tag{5.12}$$

Thus, a plot of the natural logarithms of the time required to achieve a certain Q value against the reciprocal of the absolute temperature will yield a straight line, the slope of which is dependent on the activation energy; that is, a very similar plot to Fig. 5.5 for the Del factor relationship. If both plots were superimposed on the same figure, then a continuous sterilization performance chart is obtained. The example put forward by Deindoerfer and Humphrey (1961) is shown in Fig. 5.12. Thus, in Fig. 5.12 each line of a constant Del factor specifies temperature–time regimes giving the same fractional reduction in spore number and each line of a constant nutrient quality criterion specifies temperature–time regimes giving the same destruction of nutrient. By considering the effect of nutrient destruction on product yield, limits may be imposed on Fig. 5.12 indicating the nutrient quality criterion below which no further increase in yield is achieved (i.e. the nutrient is in excess) and the nutrient quality criterion at which the product yield is at its lowest (i.e. there is no nutrient remaining). Thus, from such a plot a temperature–time regime may be chosen which gives the required Del factor and does not adversely affect the yield of the process.

The adoption of Deindoerfer and Humphrey's approach is possible only if the limiting heat-labile nutrient is identified and the Arrhenius constant and activation energy for its thermal destruction experimentally

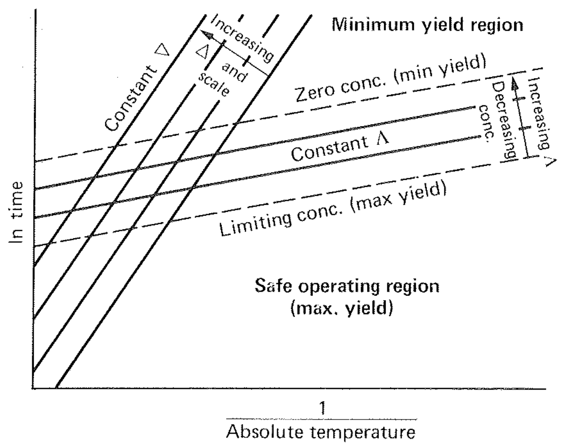

FIG. 5.12. Continuous sterilization performance chart (Deindoerfer and Humphrey, 1961).

determined. As pointed out by Banks (1979), this information may not be available for a complex fermentation medium and, therefore, the technique is fairly limited in its practical application. However, provided this information is available the technique may be used in the scale-up of continuous-sterilization processes. As discussed earlier, the Del factor is scale dependent and therefore as the volume to be sterilized is increased so the Del factor should be increased if the probability of achieving sterility is to remain the same. However, the nutrient-quality criterion is not scale dependent so that by changing the temperature–time regime to accommodate the attainment of sterility the nutrient quality may be adversely affected. Examination of Fig. 5.12 indicates that the only way in which the Del factor may be increased without any change in the nutrient quality criterion is to increase the temperature and to decrease the holding time. Although this scale-up could be performed exactly only if the thermal degradation kinetics of the medium were known, the analysis indicates that even without this information the approach would be to reduce the holding time and increase the temperature.

When designing a continuous sterilization process based on spiral heat exchangers it is important to consider the effect of suspended solids on the sterilization process. Micro-organisms contained within solid particles are given considerable protection against the sterilization treatment. If the residence time in the sterilizer is insufficient for heat to penetrate the particle then the fermentation medium may not be rendered sterile. The routine solution to this problem is to 'overdesign' the process and expose the medium to a far more severe regime than may be necessary. Armenante and Li (1993) discussed this problem in considerable detail and produced a model to predict the behaviour of a continuous system. Their analysis suggested that the temperature of the particle cores is significantly less than that of the bulk liquid. Furthermore, there is a considerable time lag in heat penetrating to the particle cores, resulting in a very different time–temperature profile for the particles as compared with the liquid medium. Thus, the temperature of the particles may not reach the critical point before they leave the sterilizer and heat penetration into the particles will continue downstream of the sterilizer. Armenante and Li's conclusion is that it is the sterilizer and/or the first cooling exchanger that should be 'overdesigned' rather than the length of the holding coil. Remember that the first cooling exchanger transfers a significant amount of heat from the sterile medium to the incoming medium and increasing its surface area would give more opportunity for the heat to penetrate the particles. This, coupled with increasing the temperature or residence time in the sterilizer, would ensure that the particle cores are up to temperature before the holding coil is reached. Also, this work suggests that it is unwise to use the direct steam injection method to heat a particulate medium because, again, there will be insufficient time for the heat to penetrate the particles.

An example of the scale up of sterilization regimes is given by the work of Jain and Buckland (1988) on the production of efrotomycin by *Nocardia lactamdurans*. In this case a beneficial interaction appeared to be occurring between the protein nitrogen source and glucose during sterilization, thus making the protein less available but resulting in a more controlled fermentation. When glucose was sterilized separately the oxygen demand of the subsequent fermentation was excessive and the fermentation terminated prematurely with very poor product formation. On scaling up the fermentation it was very difficult to attain the correct sterilization conditions using a batch regime. However, continuous sterilization using direct steam injection allowed the design of a precise process producing sterile medium with the required degree of interaction between the ingredients. The identification of this phenomenon was dependent upon careful monitoring of the small scale fermentation and consideration being given to sterilization as an important scale-up factor.

When a fermentation is scaled up it is important to appreciate that the inoculum development process is also increased in scale (see Chapter 6) and a larger seed fermenter may have to be employed to generate sufficient inoculum to start the production scale. Thus, the sterilization regime of the seed fermenter (and its medium) will also have to be scaled up. Therefore, the performance of the seed fermentation should be assessed carefully to ensure that the quality of the inoculum is maintained on the larger scale and that it has not been adversely affected by any increase in the severity of the sterilization regime.

STERILIZATION OF THE FERMENTER

If the medium is sterilized in a separate batch cooker, or is sterilized continuously, then the fermenter has to be sterilized separately before the sterile medium is added to it. This is normally achieved by heating the jacket or coils (see Chapter 7) of the fermenter with steam and sparging steam into the vessel through all entries, apart from the air outlet from which steam is allowed to exit slowly. Steam pressure is held at 15 psi in the vessel for approximately 20 minutes. It is essential that sterile air is sparged into the fermenter after the cycle is complete and a positive pressure is maintained; otherwise a vacuum may develop and unsterile air be drawn into the vessel.

STERILIZATION OF THE FEEDS

A variety of additives may be administered to a fermentation during the process and it is essential that these materials are sterile. The sterilization method depends on the nature of the additive, and the volume and feed rate at which it is administered. If the additive is fed in large quantities then continuous sterilization may be desirable, for example, Aunstrup *et al.* (1979) cited the use of continuous heat sterilization for the feed medium of microbial enzyme fermentations. Aunstrup *et al.* also referred to the use of filtration for the sterilization of certain feeds. Stowell (1987) described the use of a continuous sterilizer for the addition of oil feeds to industrial scale fermenters, but stressed that each oil has its own characteristics which makes it impossible to predict the performance of a sterilizer for different oil feeds. Batch sterilization of feed liquids normally involves steam injection into the material held in storage vessels. Whatever the sterilization system employed it is essential that all ancillary equipment and feed pipework associated with the additions are sterilizable.

STERILIZATION OF LIQUID WASTES

Process organisms which have been engineered to produce 'foreign' products and therefore contain heterologous genes are subject to strict containment regulations. Thus, waste biomass of such organisms must be sterilized before disposal. Sterilization may be achieved by either batch or continuous means but the whole process must be carried out under contained conditions. Batch sterilization involves the sparging of steam into holding tanks, whereas continuous processes would employ the type of heat exchangers which have been discussed in the previous section. A holding vessel for the batch sterilization of waste is shown in Fig. 5.13. Whichever method is employed the effluent must be cooled to below 60°C before it is discharged to waste. The sterilization processes have to be validated and are designed using the Del factor approach considered in the previous sections. However, the kinetic characteristics used in the calculations would be those of the process organism rather than of *B. stearothermophilus*. Also, the N_t value used in the design calculations would be smaller than 10^{-3} which is used for medium sterilization and would depend on the assessment of the hazard involved should the organism survive the decontamination process. Thus, the sterilization regime used for destruction of the process organism will be different from that used in sterilizing the medium.

FILTER STERILIZATION

Suspended solids may be separated from a fluid during filtration by the following mechanisms:

(i) Inertial impaction.
(ii) Diffusion.
(iii) Electrostatic attraction.
(iv) Interception.

(i) *Inertial impaction*

Suspended particles in a fluid stream have momentum. The fluid in which the particles are suspended will flow through the filter by the route of least resistance. However, the particles, because of their momentum, tend to travel in straight lines and may therefore become impacted upon the fibres where they may then remain. Inertial impaction is more significant in the filtration of gases than in the filtration of liquids.

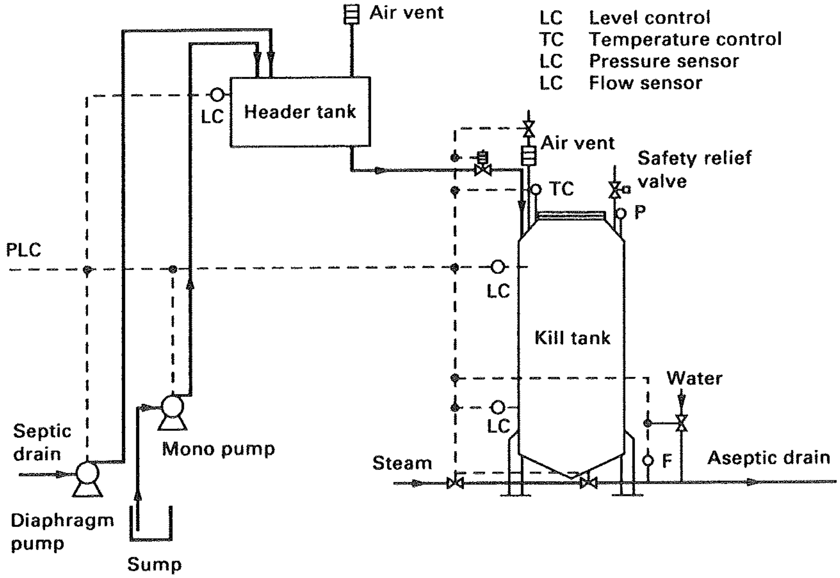

FIG. 5.13. A vessel for the batch sterilization of liquid waste from a contained fermentation (Jansson *et al.*, 1990).

(ii) *Diffusion*

Extremely small particles suspended in a fluid are subject to Brownian motion which is random movement due to collisions with fluid molecules. Thus, such small particles tend to deviate from the fluid flow pattern and may become impacted upon the filter fibres. Diffusion is more significant in the filtration of gases than in the filtration of liquids.

(iii) *Electrostatic attraction*

Charged particles may be attracted by opposite charges on the surface of the filtration medium.

(iv) *Interception*

The fibres comprising a filter are interwoven to define openings of various sizes. Particles which are larger than the filter pores are removed by direct interception. However, a significant number of particles which are smaller than the filter pores are also retained by interception. This may occur by several mechanisms — more than one particle may arrive at a pore simultaneously, an irregularly shaped particle may bridge a pore, once a particle has been trapped by a mechanism other than interception the pore may be partially occluded enabling the entrapment of smaller particles. Interception is equally important a mechanism in the filtration of gases and liquids.

Filters have been classified into two types — those in which the pores in the filter are smaller than the particles which are to be removed and those in which the pores are larger than the particles which are to be removed. The former type may be regarded as an absolute filter, so that filters of this type (provided they are not physically damaged) are claimed to be 100% efficient in removing micro-organisms. Filters of the latter type are frequently referred to as depth filters and are composed of felts, woven yarns, asbestos pads and loosely packed fibreglass. The terms absolute and depth can be misleading as they imply that absolute filtration only occurs at the surface of the filter, whereas absolute filters also have depth and filtration occurs within the filter as well as at the surface. Terms which bear more relationship to the construction of filters are 'non-fixed pore filters' (corresponding with depth filters) and 'fixed pore filters' (corresponding with absolute filters).

Non-fixed pore filters rely on the removal of particles by inertial impaction, diffusion and electrostatic attraction rather than interception. The packing material contains innumerable tortuous routes through the filter but removal is a statistical phenomenon and, thus, sterility of the product is predicted in terms of the probability of failure (similar to the situation for steam sterilization). Thus, in theory, the removal of micro-organisms by a fibrous filter cannot be absolute as there is always the possibility of an organism passing through the filter, regardless of the filter's depth. Also, because the fibres are not cemented into position an increase in the pressure to which the filter is subjected may result in movement of the material, producing larger channels through the filter. Increased pressure may also result in the displacement of previously trapped particles.

Fixed pore filters are constructed so that the filtration medium will not be distorted during operation so that the flow patterns through the filter will not change due to disruption of the material. Pore size is controlled during manufacture so that an absolute rating can be quoted for the filter — i.e. the removal of particles above a certain size can be guaranteed. Thus, interception is the major mechanism by which particles are removed. Because fixed pore filters have depth they are also capable of removing particles which are smaller than the pores by the mechanisms of inertial impaction, diffusion and attraction and these mechanisms do play significant roles in the filtration of gases. Fixed pore filters are superior for most purposes in that they have absolute ratings, are less susceptible to changes in pressure and are less likely to release trapped particles. The major disadvantage associated with absolute filters was the resistance to flow they present and, hence, the large pressure drop across the filters which represents a major operating cost. However, modern absolute filter cartridges which have been developed by many filtration companies contain pleated membranes with very large surface areas, thereby minimizing the pressure drop across the filter.

It is important to realize that filters should be steam sterilized before and after operation (see Chapter 7). Thus, the materials must be stable at high temperatures and the steam must be free of particulate matter because the filter modules are particularly vulnerable to damage at high temperatures. Thus, the steam itself is filtered through stainless steel mesh filters rated at 1 μm.

Filter sterilization of fermentation media

Media for animal-cell culture cannot be sterilized by steam because they contain heat-labile proteins. Thus, filtration is the method of choice and as may be appreciated from the previous discussion, fixed pore or absolute filtration is the better system to use. An ideal filtration system for the sterilization of animal cell culture media must fulfil the following criteria:

(i) The filtered medium must be free of fungal, bacterial and mycoplasma contamination.
(ii) There should be minimal adsorption of protein to the filter surface.
(iii) The filtered medium should be free of viruses.
(iv) The filtered medium should be free of endotoxins.

Several filter manufacturers now supply absolute filtration systems for the sterilization of animal cell culture medium. Such systems consist of membrane cartridges which are fitted into stainless steel, steam sterilizable modules. The membranes for media filtration are constructed from steam sterilizable hydrophilic material and are treated to produce a filtrate of particular quality. For example, if minimal protein adsorption is a major criterion then a specially coated filter membrane is used. It would be very difficult to construct a single filtration membrane which would fulfil all four criteria cited above. Thus, a series of filters are used to achieve the desired result. The example shown in Fig. 5.14 is provided by Pall Process Filtration Ltd and illustrates a system to produce sterile, mycoplasma free serum and consists of four filters arranged in sequence. The first filter is a positively charged polypropylene pre-filter with an absolute rating of 5 μm for the removal of coarse precipitates, clot-like material and other gross contaminants; the second filter is also positively charged polypropylene but with an absolute rating of 0.5 μm for bulk microbial removal, deformable gels, lipid-based materials and endotoxin reduction; the third filter is a single layered, nylon/polyester positively charged filter with a 0.1-μm absolute rating for further microbial and endotoxin removal and optimum protection of the final filter; the fourth filter is similar to the third and has the same rating, but is double layered and removes mycoplasmas, gives absolute sterility and final endotoxin control. Thus, the combination of four filters gives a sequential removal of decreasingly small particles and prolongs the life of the final filter. If it is necessary to remove viral contamination then a final 0.04-μm nylon/polyester filter would be added.

Similar systems may be used in downstream processing of animal cell products where the rating and properties of the filters would be optimized for the particular process. Figure 5.15 illustrates a system for the removal of cells and cell debris from an animal-cell fermentation broth. The pre-filter is a polypropylene 1.0-μm rated filter to remove the bulk of the cells and debris and the second filter is an hydroxyl modified nylon/polyester 0.2-μm rated filter giving absolute cell removal with minimal protein adsorption.

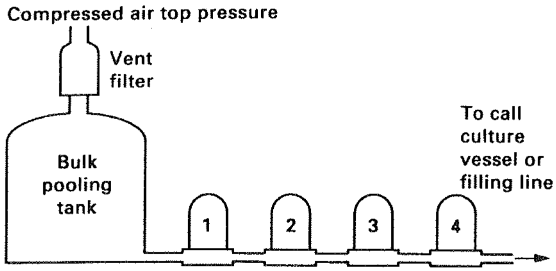

FIG. 5.14. Filtration system for the provision of sterile, mycoplasma free serum.
Filter 1. 5μm absolute rated prefilter for removal of coarse precipitates.
Filter 2. 0.5μm absolute rated prefilter for bulk bioburden removal.
Filter 3. 0.1μm absolute rated single layer prefilter for futher bioburden and endotoxin removal.
Filter 4. 0.1μm absolute rated double layer final filter for absolute sterility, mycoplasma removal and further endotoxin control.
(Pall Process Filtration Ltd., Portsmouth, U.K.)

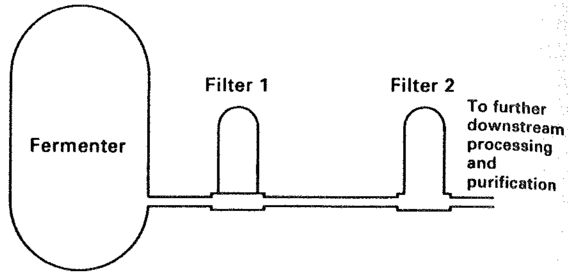

FIG. 5.15. Filtration system for the removal of cells and cell debris from an animal cell culture fermentation.
Filter 1. 1.0μm absolute rated prefilter for bulk cell and cell debris removal.
Filter 2. 0.2μm absolute rated single layer 'Bio-Inert' filter for final bioburden removal.
(Pall Process Filtration Ltd., Portsmouth, U.K.)

Filter sterilization of air

Aerobic fermentations require the continuous addition of considerable quantities of sterile air (see Chapter 9). Although it is possible to sterilize air by heat treatment, the most commonly used sterilization process is filtration. Fixed pore filters (which have an absolute rating) are very widely used in the fermentation industry and several manufacturers produce filtration systems for air sterilization. These systems, like those for the sterilization of liquids, consist of pleated membrane cartridges designed to be accommodated in stainless steel modules. A selection of such cartridges and holders is shown in Fig. 5.16 and a sectioned filter unit is shown in Fig. 5.17. The most common construction material used for for the pleated membranes for air sterilization is PTFE, which is hydrophobic and is therefore resistant to wetting. Also, PTFE filters may be steam sterilized and are resistant to ammonia which may be injected into the air stream, prior to the filter, for pH control. As was seen for the filter sterilization of liquids it is essential that a prefilter is incorporated up-stream of the absolute filter. The prefilter traps large particles such as dust, oil and carbon (from the compressor) and pipescale and rust (from the pipework). The use of a coalescing prefilter also ensures the removal of water from the air; entrained water is coalesced in the filter (air flow being from the inside of the filter to the outside) and is discharged via an automatic drain. Figure 5.18 illustrates the layout of such a filtration unit showing the steam sterilization system and Fig. 5.19 is a photograph of the air sterilization plant for an 85 m^3 fermenter. Smith (1981) cited the use of absolute filtration in the sterilization of air for the ICI biomass continuous fermenter.

Sterilization of fermenter exhaust air

In many traditional fermentations the exhaust gas from the fermenter was vented without sterilization or vented through relatively inefficient depth filters. With the advent of the use of recombinant organisms and a greater awareness of safety and emission levels of allergic compounds the containment of exhaust air is more common (and in the case of recombinant organisms, compulsory). Fixed pore membrane modules are also used for this application but the system must be able to cope with the sterilization of water saturated air, at a relatively high temperature and carrying a large contamination level. Also, foam may overflow from the fermenter into the air exhaust line. Thus, some form of pretreatment of the exhaust gas is necessary before it enters the absolute filter. This pretreatment may be a hydrophobic prefilter or a mechanical separator to remove water, aerosol particles and foam. The pretreated air is then fed to a 0.2-μm hydrophobic filter. Again, it is important to appreciate that the filtration system must be steam sterilizable. Figures 5.18 and 5.19 illustrate the prefilter and mechanical separator systems respectively.

The theory of depth filters

Although most fermentation companies rely upon the pleated membrane, fixed pore (absolute rated) filter

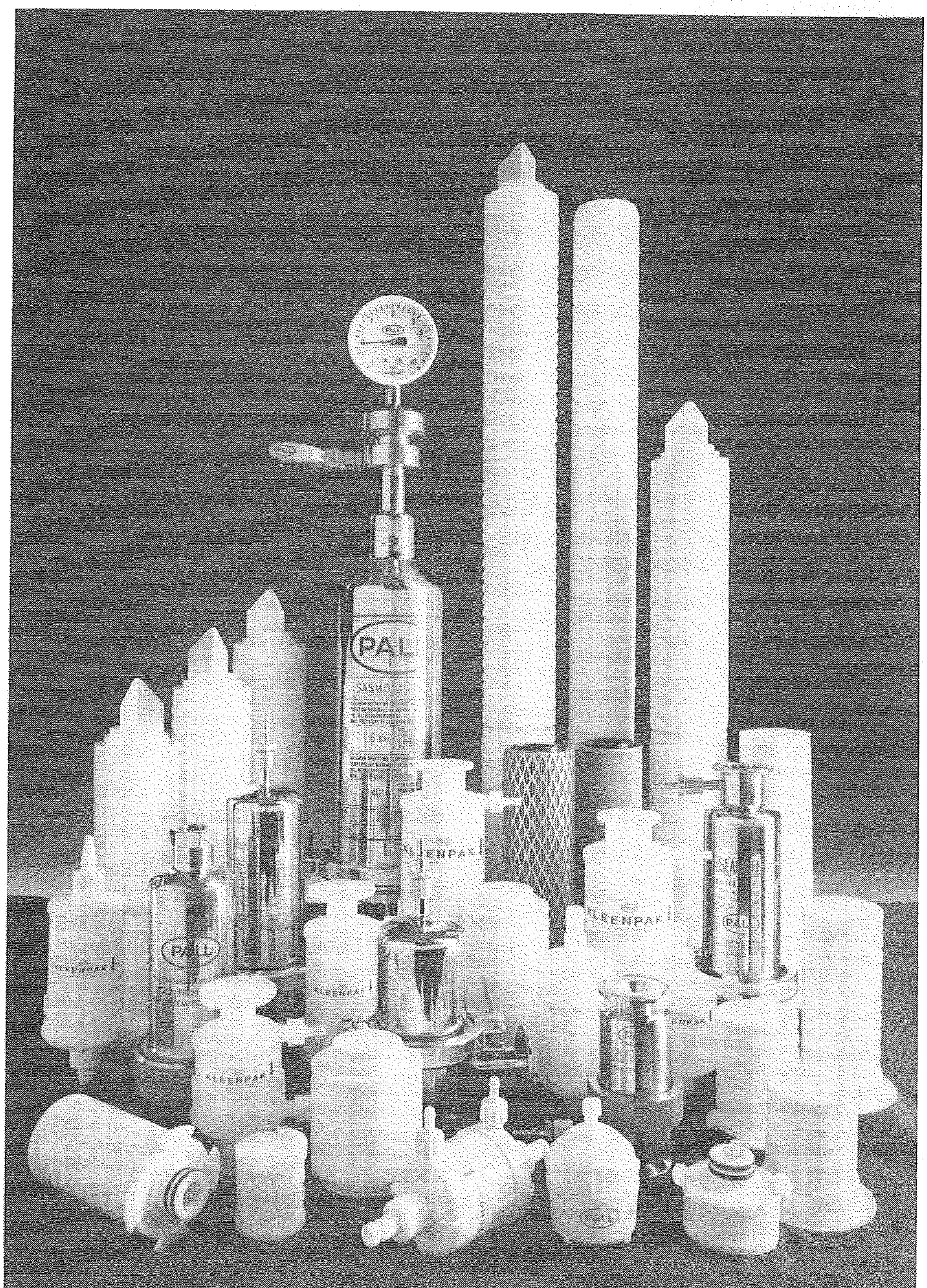

FIG. 5.16. A selection of absolute membrane filter cartridges and stainless steel housings (Pall Process Filtration, Portsmouth, U.K.).

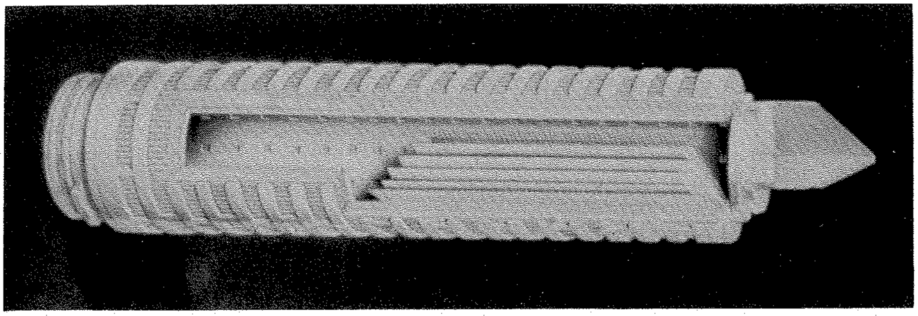

FIG. 5.17. An absolute membrane filter sectioned to show the pleated membrane structure (Pall Process Filtration, Portsmouth, U.K.).

systems it is necessary to consider the theory of depth filtration. Aiba *et al.* (1973) have given detailed quantitative analysis of these mechanisms but this account will be limited to a description of the overall efficiency of operation of fibrous filters. Several workers (Ranz and Wong, 1952; Chen, 1955) have put forward equations relating the collection efficiency of a filter bed to various characteristics of the filter and its components. However, a simpler description cited by Richards (1967) may be used to illustrate the basic principles of filter design.

If it is assumed that if a particle touches a fibre it remains attached to it, and that there is a uniform concentration of particles at any given depth in the filter, then each layer of a unit thickness of the filter should reduce the population entering it by the same proportion; which may be expressed mathematically as:

$$dN/dx = -KN \qquad (5.13)$$

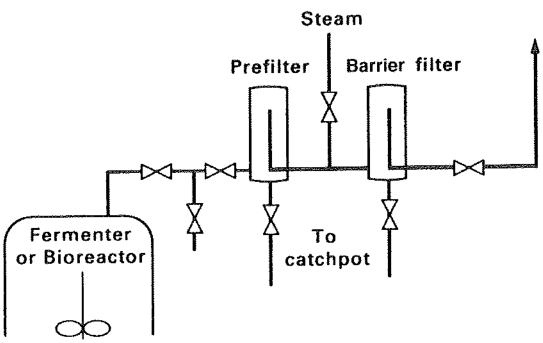

FIG. 5.18. Dual hydrophobic filter system for the sterilization of off-gas from a fermenter (Pall Process Filtration Ltd., Portsmouth, U.K.).

where N is the concentration of particles in the air at a depth, x, in the filter and
K is a constant.

On integrating equation (5.13) over the length of the filter it becomes:

$$N / N_0 = e^{-Kx} \qquad (5.14)$$

where N_0 is the number of particles entering the filter
and N is the number of particles leaving the filter.

On taking natural logarithms, equation (5.14) becomes:

$$\ln (N / N_0) = -Kx. \qquad (5.15)$$

Equation (5.15) is termed the log penetration relationship. The efficiency of the filter is given by the ratio of the number of particles removed to the original number present, thus:

$$E = (N_0 - N)/N_0 \qquad (5.16)$$

where E is the efficiency of the filter.
But:

$$(N_0 - N)/N_0 = 1 - (N / N_0). \qquad (5.17)$$

Substituting $N / N_0 = e^{-Kx}$

Thus:

$$(N_0 - N)/N_0 = 1 - e^{-Kx} \qquad (5.18)$$

and $E = 1 - e^{-Kx}$.

The log penetration relationship [equation (5.15)] has been used by Humphrey and Gaden (1955) in filter design, by using the concept X_{90}, the depth of filter required to remove 90% of the total number of particles entering the filter; thus:

If N_0 were 10 and x were X_{90}, then N would be 1:

$$\ln (1/10) = -KX_{90}$$

or

$$2.303 \log_{10}(1/10) = -KX_{90}$$

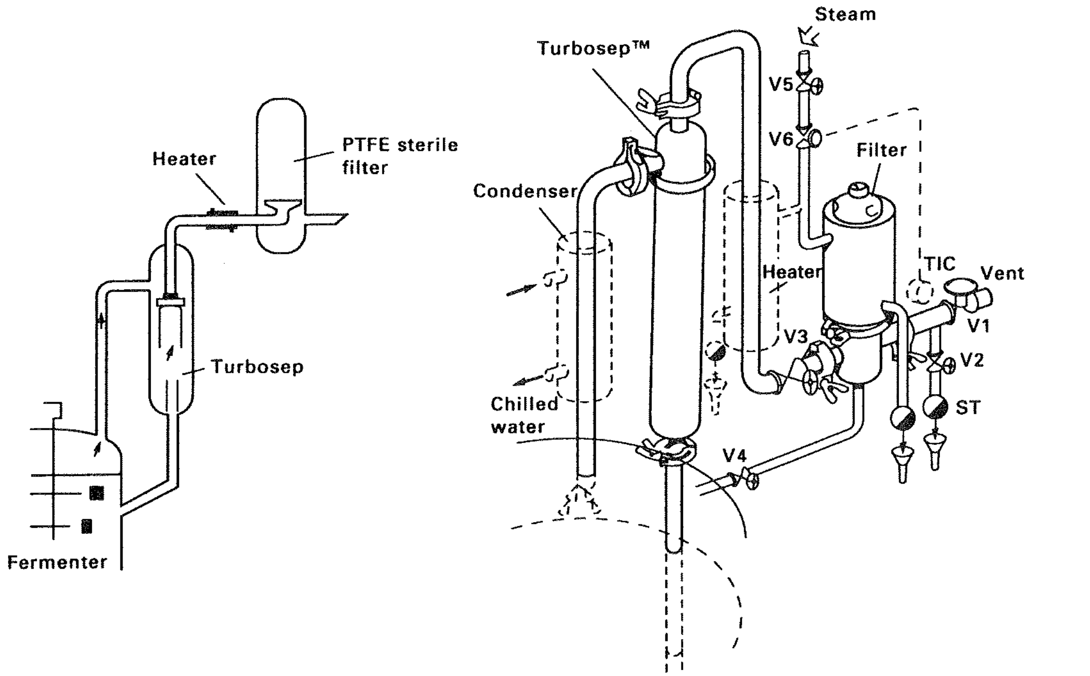

FIG. 5.19. A mechanical separator and hydrophobic filter system for the sterilization of off-gas from a fermenter. Left. Cut-away diagram. Right. Equipment arrangement, showing steam supply. V1–V6, valves; O, steam, traps (Domnick Hunter Ltd., Birtley, Co. Durham, U.K.).

$$2.303(-1) = -KX_{90},$$

therefore, $$X_{90} = 2.303/K. \qquad (5.19)$$

Consideration of equation (5.15) indicates that a plot of the natural logarithm of N/N_0 against x, filter length, will yield a straight line of slope K. To obtain the information for a plot of this type, data must be gleaned for the removal of organisms from an air stream by filters of increasing length which would involve assessment of microbial levels in the air entering and leaving the filter. Humphrey and Gaden (1955) and Richards (1967) described equipment with which such assessments could be recorded.

The value of K is affected by the nature of the filter material and by the linear velocity of the air passing through the filter. Figure 5.20 is a typical plot of K and X_{90} against linear air velocity from which it may be seen that K increases to an optimum with increasing air velocity, after which any further increase in air velocity results in a decrease in K. Table 5.3 (Riviere, 1977) summarizes the effects of linear air velocity on the removal of a range of micro-organisms with a variety of filter materials.

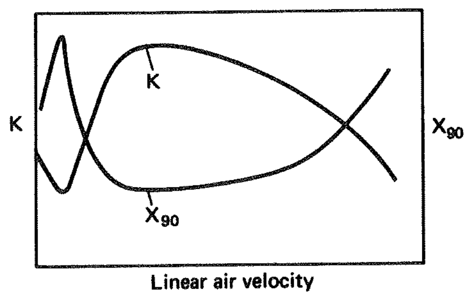

FIG. 5.20. The effect of increasing linear air velocity on K and X_{90} of a filtration system (Richards, 1967).

The increase in K with increasing air velocity is probably due to increased impaction, illustrating the important contribution this mechanism makes to the removal of organisms. The decrease in K values at high air velocities is probably due to disruption of the filter, allowing channels to develop and fibres to vibrate, resulting in the release of previously captured organisms.

TABLE 5.3. *X_{90} values for the removal of a range of micro-organisms by a variety of filtration materials* (Humphrey, 1960; Rivierre, 1977)

Filter material	Diameter of fibres (μm)	Micro-organism	Air speed (cm s^{-1})	X_{90} (cm)
Glass wool	16	*Bacillus subtilis* spores	3	4
			15	9
			30	11.5
			150	1.5
			300	0.4
	18.5	*Serratia marcescens*	3.9	3.1
Glass fibre	8.5	*Escherichia coli*	3	0.4
			15	0.6
			30	0.7
			150	0.8
			300	1.1
Norite (15–30 mesh)	—	*Bacillus cereus* spores	1.4	1.7
			6.4	1.5
Activated carbon (4–8 mesh)		*Bacillus cereus* spores	18	1.7
			28.5	8.7

The design of depth filters

Equation (5.15), the log penetration relationship, is the same form as equation (5.3) in the derivation of thermal-death kinetics. In the case of heat sterilization the theory predicts that an infinite time is required to reduce the population to zero, whereas the theory of filtration predicts that a filter of infinite length is required to remove all organisms from an air stream. Thus, it is not surprising that the same approach is adopted in the design of filters and heat-sterilization cycles, in that an acceptable probability of contamination is determined. The probability of one fermentation in a thousand being contaminated is frequently used in filter design, as it is in the design of heat-sterilization cycles. Having arrived at an acceptable probability of contamination and determined the filtration characteristics (i.e. the value of K) of the material to be used, a filter may be designed to filter a certain volume of air containing a certain number of organisms; the following example illustrates the design calculation approach used by Richards (1967):

It is required to provide a 20-m^3 fermenter with air at a rate of 10 m^3 min^{-1} for a fermentation lasting 100 hours. From an investigation of the filter material to be used, the optimum linear air velocity was shown to be 0.15 m sec^{-1}, at which the value of K was 1.535 cm^{-1}. The dimensions of the filter may be calculated as follows:

The log penetration relationship states that:

$$\ln(N/N_0) = -Kx.$$

The air in the fermentation plant contained approximately 200 micro-organisms m^{-3}.
Therefore,

$$N_0 = \text{total amount of air provided} \times 200,$$

$$N_0 = 10 \times 60 \times 100 \times 200$$

$$= 12 \times 10^6 \text{ organisms}.$$

The acceptable degree of contamination is one in a thousand,
therefore $N = 10^{-3}$,

$$\ln\{10^{-3}/(12 \times 10^6)\} = -Kx,$$

$$\ln 8.33 \times 10^{-11} = -Kx,$$

$$\ln 8.33 \times 10^{-11} = -1.535x,$$

$$x = -23.21/-1.535 = 15.12 \text{ cm}.$$

Therefore, the filter to be used should be 15.12 cm long.
The cross-sectional area of the filter is given by the volumetric air flow rate divided by the linear air velocity:

$$\pi r^2 = 10/0.15 \times 60$$

where r is the radius of the filter

$$r = 0.59 \text{ m}.$$

Thus the filter to be employed should be 15.12 cm long and 0.59 m radius.

However, as Humphrey (1960) pointed out, the efficient operation of the filter is dependent on the supply of air at the optimum linear velocity. If the air velocity increases or decreases the value of K will decrease, resulting in a loss of filtration efficiency. Considering the example calculation, if the linear air velocity were to drop to 0.03 m sec^{-1}, then the value of K would decline to 0.2 cm^{-1}. The number of organisms which would enter the fermentation in 1 minute at this reduced air-flow rate would be as calculated below:

$$\ln (N/N_0) = -Kx.$$

At a linear air velocity of 0.03 m sec^{-1}, in 1 minute 0.03 $\times$ 60 $\times$ the cross-sectional area of the filter m^{-3} of air would enter the filter, i.e. 1.98 m^3. At a microbial contamination level of 200 organisms m^{-3} this means that 396 organisms would enter the filter in 1 minute. Thus:

$$\ln (N/396) = -0.2 \times 15.12,$$

$$N = 19.24.$$

Therefore, 19.24 organisms would have entered the fermenter in 1 minute at the decreased air-flow rate. If the filter had been designed to meet this contingency, then the length would have been:

$$\ln (10^{-3}/396) = -0.2x,$$

$$x = 64.4.$$

Thus a filter length of 64.4 cm would have been required to have maintained the same probability of contamination over the 1 minute of reduced air flow.

This example illustrates the hazards of attempting very precise design and the necessity to consider the reliability of ancillary equipment in any design calculation.

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CHAPTER 6

The Development of Inocula for Industrial Fermentations

INTRODUCTION

IT IS ESSENTIAL that the culture used to inoculate a fermentation satisfies the following criteria:

1. It must be in a healthy, active state thus minimizing the length of the lag phase in the subsequent fermentation.
2. It must be available in sufficiently large volumes to provide an inoculum of optimum size.
3. It must be in a suitable morphological form.
4. It must be free of contamination.
5. It must retain its product-forming capabilities.

The process adopted to produce an inoculum meeting these criteria is called inoculum development. Hockenhull is credited with the quotation "once a fermentation has been started it can be made worse but not better" (Calam, 1976). Whereas this is an over-statement it does illustrate the importance of inoculum development. Much of the variation observed in small-scale laboratory fermentations is due to poor inocula being used and, thus, it is essential to appreciate that the establishment of an effective inoculum development programme is equally important regardless of the scale of the fermentation. Such a programme not only aids consistency on a small scale but is invaluable in scaling up the fermentation and forms an essential part in progressing a new process.

A critical factor in obtaining a suitable inoculum is the choice of the culture medium. It must be stressed that the suitability of an inoculum medium is determined by the subsequent performance of the inoculum in the production stage. As discussed elsewhere (Chapter 4), the design of a production medium is determined not only by the nutritional requirements of the organism, but also by the requirements for maximum product formation. The formation of product in the seed culture is not an objective during inoculum development so that the seed medium may be of a different composition from the production medium. However, Lincoln (1960) stated that the lag time in a fermentation is minimized by growing the culture in the 'final-type' medium. Lincoln's argument is an important one, so the inoculum development medium should be sufficiently similar to the production medium to minimize any period of adaptation of the culture to the production medium, thus reducing the lag phase and the fermentation time. Furthermore, Hockenhull (1980) pointed out the dangers of using very different media in consecutive stages. Major differences in pH, osmotic pressure and anion composition may result in very sudden changes in uptake rates which, in turn, may affect viability. Hockenhull also emphasized that for antibiotic fermentations the inoculum medium should contain sufficient carbon and nitrogen to support maximum growth until transfer, so that secondary metabolism remains repressed during growth of the inoculum. If secondary metabolism is derepressed in the seed fermentation, then selection may enrich the culture with non-producing variants having a growth advantage over high-producing types. Hockenhull drew attention to Righelato's (1976) work in which it was shown that chemostat culture of *Penicillium chrysogenum* under carbohydrate-limited conditions led to a loss of penicillin synthesizing ability and an increase in the proportion of non-conidiated variants whereas this did not occur in ammonia-, phosphate- or sulphate-limited conditions. The relevance of this phenomenon is supported by Hockenhull's observation that *P. chrysogenum* inocula produced under non-limiting conditions are remarkably free from variants whereas variants arise relatively frequently during the carbon-limited production phase. Examples of inoculum and production media are given in Table 6.1, from which it may be seen that inoculum media are, generally, less nutritious than production media and contain a lower level of carbon.

The quantity of inoculum normally used is between 3 and 10% of the medium volume (Lincoln, 1960; Meyrath and Suchanek, 1972; Hunt and Stieber, 1986). A relatively large inoculum volume is used to minimize the length of the lag phase and to generate the maximum biomass in the production fermenter in as short a time as possible, thus increasing vessel productivity. Thus, starting from a stock culture, the inoculum must be built up in a number of stages to produce sufficient biomass to inoculate the production-stage fermenter. This may involve two or three stages in shake flasks and one to three stages in fermenters, depending on the size of the ultimate vessel. Throughout this procedure there is a risk of contamination and strain degeneration necessitating stringent quality-control procedures. The greater the number of stages between the master culture and the production fermenter the greater the risk of contamination and strain degeneration. Therefore, a compromise may be reached regarding the size of the inoculum to be used and the risk of contamination and strain degeneration. Another factor to be considered in the determination of the inoculum volume is the economics of the process. A seed fermenter 10% of the size of the production fermenter represents a considerable financial investment and must be justified in terms of productivity. A large-scale continuous fermentation for the production of biomass

TABLE 6.1. *Inoculum development and production media for a range of processes*

Process	Inoculum development medium		Production medium		References
Griseofulvin	Whey powder, Lactose } to give:		Lactose	7%	Rhodes *et al.* (1957)
	Lactose	3.5%	Corn-steep liquor solids to give:		
	Nitrogen	0.05%	Nitrogen	0.2%	
	Corn-steep liquor solids (to give approx. 0.04% N)	0.38%	Limestone	0.8%	
			KH_2PO_4	0.4%	
			KCl	0.1%	
	KH_2PO_4	0.4%			
	KCl	0.05%			
Clavulanic acid	Soybean flour	1.0%	Soybean flour	1.5%	Butterworth (1984)
	Dextrin	2.0%	Oil	1.0%	
	Pluronic L81 (Antifoam)	0.03%	KH_2PO_4	0.1%	
		($g\ dm^{-3}$)		($g\ dm^{-3}$)	
Vitamin B_{12}	Sugar beet Molasses	70		105	Spalla *et al.* (1989)
	Sucrose	—		15	
	Betaine	—		3	
	$NH_4H_2PO_4$	0.8		—	
	$(NH_4)_2SO_4$	2		2.5	
	$MgSO_4$	0.2		0.2	
	$ZnSO_4$	0.02		0.08	
	5-6 Dimethyl-benzimidazole	0.005		0.025	
Lysine	Cane molasses	5%		20%	Nakayama (1972)
	Corn-steep liquor	1%		—	
	$CaCO_3$	1%		—	
	Soybean meal hydrolysate	—		1.8%	

would be expected to operate at steady state in excess of 100 days. Thus, the number of times that the fermenter is inoculated should be very few compared with batch or fed-batch systems. In such circumstances it may be more economic to compromise on the size of the inoculum and to tolerate a relatively lengthy period of growth up to maximum biomass than to invest a large seed vessel which would be used on very few occasions. This is particularly relevant for biomass continuous processes where one very large fermenter may be used and, thus, any seed vessel would only be servicing the one production vessel.

A typical inoculum-development programme will now be described in detail. The master culture (see Chapter 3 for an account of master-culture maintenance) is reconstituted and plated on to solid medium; approximately ten colonies of typical morphology of high producers are selected and inoculated on to slopes as the sub-master cultures, each sub-master culture being used for a new production run. At this stage, shake flasks may be inoculated to check the productivity of these cultures, the results of such tests being known before the developing inoculum eventually reaches the production plant. A sub-master culture is used to inoculate a shake flask (250 or 500 cm^3 containing 50 or 100 cm^3 medium) which, in turn, is used as inoculum for a larger flask, or a laboratory fermenter, which may then be used to inoculate a pilot-scale fermenter. Culture purity checks are carried out at each stage to detect contamination as early as possible. Although the results of these tests may not be available before the culture has reached the production plant, at least it is known at which stage in the procedure contamination occurred. For a sporulating organism the process may be modified to facilitate the use of a spore suspension as inoculum and this will be discussed in more detail later in this chapter.

Lincoln (1960) suggested a more elaborate procedure for the development of inoculum for bacterial fermentations which, with minor modifications, is applicable to any type of culture. The procedure involved the use of one sub-master culture to develop a bulk inoculum which was subdivided, stored in a frozen state and used as inocula for several months. A single colony, derived from a sub-master culture, was inoculated into liquid medium and grown to maximum log phase. This culture was then transferred into nineteen times its volume of medium and incubated again to the maximum log phase, at which point it was dispensed in 20-cm^3 volumes, plug frozen and stored at below $-20°$. At least 3% of the samples were tested for purity and productivity in subsequent fermentation and, provided these were suitable, the remaining samples could be used as initial inocula for subsequent fermentations. To use one of the stored samples as inoculum it was thawed and used as a 5% inoculum for a seed culture which, in turn, was used as a 5% inoculum for the next stage in the programme. This procedure ensured that a proven inoculum was used for the penultimate stage in inoculum development.

CRITERIA FOR THE TRANSFER OF INOCULUM

The physiological condition of the inoculum when it is transferred to the next culture stage can have a major effect on the performance of the fermentation. The optimum time of transfer must be determined experimentally and then procedures established so that inoculation with an ideal culture may be achieved routinely. These procedures include the standardization of cultural conditions and monitoring the state of an inoculum culture so that it is transferred at the optimum time, i.e. in the correct physiological state. The most widely used criterion for the transfer of vegetative inocula is biomass and such parameters as packed cell volume, dry weight, wet weight, turbidity, respiration, residual nutrient concentration and morphological form have been used (Hockenhull, 1980; Hunt and Stieber, 1986). Ettler (1992) demonstrated that the rheological behaviour of *Streptomyces noursei* could be used as the transfer criterion in the nystatin fermentation. The rheology of the seed fermentation changed from Newtonian to non-Newtonian behaviour and the optimum inoculum transfer time corresponded with this transformation.

Criteria which may be monitored on-line are the most convenient parameters to use as indicators of inoculum quality and these would include dissolved oxygen, pH (although pH would normally be controlled in seed fermentations) and oxygen or carbon dioxide in the effluent gas. Parton and Willis (1990) advocated the use of the carbon dioxide production rate (CPR) as a transfer criterion which requires analysis of the fermenter effluent air (see Chapter 8). This approach is suitable only when transfer is being made from a fermenter, but Parton and Willis stressed the importance of adopting this strategy even for the inoculation of laboratory-scale fermentations despite the fact that an adequate inoculum volume could be produced in shake flask. These workers provide an excellent example of the effect of inoculum transfer time on the productivity of a streptomycete secondary metabolite, as shown in Fig. 6.1a, b and c. The CPR of the

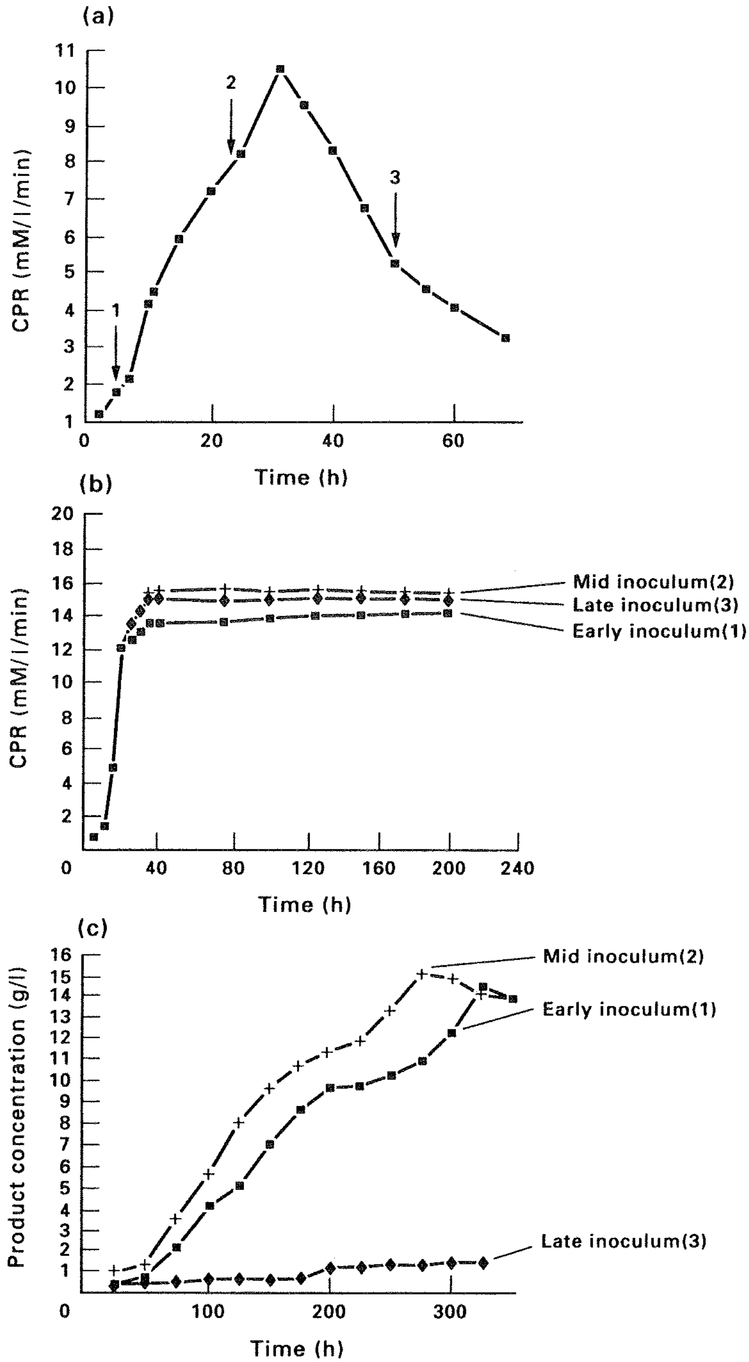

FIG. 6.1. The effect of inoculum age on growth and productivity in a streptomycete fermentation. (a) The carbon dioxide production rate (CPR) profile of the inoculum culture showing the points (1, 2 and 3) at which inocula were removed. (b) The effect of inoculum age on the CPR of the production fermentation. (c) The effect of inoculum age on productivity in the production fermentation (Parton and Willis, 1990).

inoculum fermentation and the points at which inoculum was transferred are shown in Fig. 6.1a. The CPR of the subsequent production fermentations are shown in Fig. 6.1b, from which it may be seen that the three fermentations performed similarly. However, Fig. 6.1c illustrates the very different secondary metabolite production of the three fermentations. Thus, although the time of transfer had only a marginal influence on biomass in the production fermentation the effect on product formation was critical. It should be emphasized that the amount of biomass transferred was standardized for the three fermentations and, thus, the differences in performance were due to the physiological states of the inocula.

In recent years, probes have been developed for on-line assessment of biomass (see Chapter 8) and these could be invaluable in estimating the time of inoculum transfer. Boulton *et al.* (1989) reported the use of a biomass sensor (the Bugmeter) to control the yeast pitching rate (inoculum level) in brewing. The probe measures the dielectric permittivity of viable yeast cells and is unaffected by the presence of dead cells, air bubbles or detritus, making it ideal for the routine monitoring of yeast inoculum. Using the probe, these workers developed an automatic inoculum dispenser allowing a preset viable yeast mass to be transferred from a yeast storage vessel to the brewery fermentation.

Alford *et al.* (1992) reported the use of a real-time expert computer system to predict the time of inoculum transfer for industrial-scale fermentations. The system involves the comparison of on-line fermentation data with detailed historical data of the process. A problem with the interpretation of carbon dioxide production rate figures is that the data are not available continuously because the analyser is not dedicated to any one fermenter, but is analysing process streams from a large number of vessels via a multiplexer system (see Chapter 8). Thus, a fermentation may have passed a critical stage between monitoring times. Also, occasional false readings may be generated. The expert system enabled the verification of data points as well as prediction of the outcome of the fermentation from early information. Data from seed fermentations were analysed by the expert system and the transfer time predicted. As a result of this approach operators were able to plan their work more effectively, the need for manual sampling was reduced and early warning of contamination was provided if the seed-culture profile predicted from early readings was abnormal.

Smith and Calam (1980) compared the quality and enzymic profile of differently prepared inocula *Penicillium patulum* (producing griseofulvin) and demonstrated that a low level of glucose 6-phosphate dehydrogenase was indicative of a good quality inoculum. The enzyme profile of good quality inoculum was established early in the growth of the seed culture. Thus, this approach could be used to assess the cultural conditions giving rise to satisfactory inoculum, but would be of less value in determining the time of transfer.

Yeast, unicellular bacterial, fungal and streptomycete fermentations have different requirements for inoculum development and these are dealt with separately.

THE DEVELOPMENT OF INOCULA FOR YEAST PROCESSES

Whilst the largest industrial fermentations utilizing yeasts are the brewing of beer and the production of biomass, recent processes have also been established for the production of recombinant products.

Brewing

It is common practice in the British brewing industry to use the yeast from the previous fermentation to inoculate a fresh batch of wort. The brewing terms used to describe this process are 'crop', referring to the harvested yeast from the previous fermentation, and 'pitch', meaning to inoculate. One of the major factors contributing to the continuation of this practice is the wort-based excise laws in the United Kingdom where duty is charged on the sugar consumed rather than the alcohol produced. Thus, dedicated yeast propagation systems are expensive to operate because duty is charged on the sugar consumed by the yeast during growth. It can then be appreciated that the reduced cost of using yeast from a previous fermentation is an attractive proposition (Boulton, 1991). The dangers inherent in this practice are the introduction of contaminants and the degeneration of the strain, the most common degenerations being a change in the degree of flocculence and attenuating abilities of the yeast. In breweries employing top fermentations in open fermenters these dangers are minimized by collecting yeast to be used for future pitching from 'middle skimmings'. During the fermentation the yeast cells flocculate and float to the surface, the first cells to do this being the most flocculent and the last cells the least flocculent. As the head of yeast develops, the